

Task1

```
package Day24;
public class Apple { no usages
    private Apple() { no usages
        // private constructor to prevent instantiation
    }

    public static Mobile getMobile(String model) { no usages
        if ("iphone16".equalsIgnoreCase(model)) {
            return new Mobile(desc: "Here is your iPhone 16");
        } else if ("iphone16MaxPro".equalsIgnoreCase(model)) {
            return new Mobile(desc: "Here is your iPhone 16 Max Pro");
        } else {
            return new NoMobile();
        }
    }
}
```

"D:\Program Files\Java\jdk-24\bin\java
Here are your Apple models:
Here is your iPhone 16
Thank you for visiting Mobile Store!
Process finished with exit code 0

```
package Day24;
public class Mobile { 3 usages 1 inheritor
    String desc; 2 usages

    public Mobile(String desc) { 2 usages
        this.desc = desc;
    }

    public void getDesc() { no usages
        System.out.println(desc);
    }
}
```

```

package Day24;
public class MobileStore {  no usages

    private MobileStore() {  no usages
        System.out.println("Hello, welcome to the world of Mobile");
    }

    public static Mobile getMobile(String brand, String model) {  no usages
        if ("Apple".equalsIgnoreCase(brand)) {
            System.out.println("Here are your Apple models:");
            return Apple.getMobile(model);
        }
        // Future expansion for other brands like Samsung can go here.
        // else if ("Samsung".equalsIgnoreCase(brand)) {
        //     return Samsung.getMobile(model);
        // }

        return new NoMobile(); // unknown brand
    }
}

```

```

package Day24;
public class NoMobile extends Mobile {  1 usage
    public NoMobile() {  1 usage
        super(desc: "Sorry, invalid model.");
    }
}

```

```

package Day24;

public class ClientAbstractFactory {
    public static void main(String[] args) {
        Mobile mObj = MobileStore.getMobile(brand: "Apple", model: "iphone16");
        mObj.getDesc();

        System.out.println("Thank you for visiting Mobile Store!");
    }
}

```

Task2

```

package Day24;
// product Component
class Laptop { 1 usage
    private int memory; 3 usages
    private int storage; 3 usages

    public Laptop() {} no usages

    public int getMemory() { no usages
        return memory;
    }

    public void setMemory(int memory) { no usages
        this.memory = memory;
    }

    public int getStorage() { no usages
        return storage;
    }

    public void setStorage(int storage) { no usages
        this.storage = storage;
    }
}

```

```
@Override  
public String toString() {  
    return "Laptop [Memory=" + memory + "GB, Storage=" + storage + "GB]";  
}
```

```
package Day24;  
  
public interface LaptopBuilder { 8 usages 1 implementation  
    LaptopBuilder buildMemory(int memory); 1 usage 1 implementation  
    LaptopBuilder buildStorage(int storage); 1 usage 1 implementation  
    Laptop build(); 1 usage 1 implementation  
}
```

```
package Day24;

public class LaptopConcreteBuilder implements LaptopBuilder { 1 usage
    private Laptop laptop; 4 usages

    public LaptopConcreteBuilder() { 1 usage
        this.laptop = new Laptop();
    }

    @Override 1 usage
    public LaptopBuilder buildMemory(int memory) {
        laptop.setMemory(memory);
        return this;
    }

    @Override 1 usage
    public LaptopBuilder buildStorage(int storage) {
        laptop.setStorage(storage);
        return this;
    }

    @Override 1 usage
    public Laptop build() {
```

```

package Day24;

//Director (optional)
public class LaptopDirector { 2 usages
    private LaptopBuilder laptopBuilder; 2 usages

    public LaptopDirector(LaptopBuilder laptopBuilder) { 1 usage
        this.laptopBuilder = laptopBuilder;
    }

    public Laptop constructLaptop() { 1 usage
        return laptopBuilder
            .buildMemory(16) // Example: 16GB RAM
            .buildStorage(512) // Example: 512GB SSD
            .build();
    }
}

```

```

package Day24;
public class ClientBuildMethodDP {
    public static void main(String[] args) {
        LaptopBuilder lbobj = new LaptopConcreteBuilder();
        LaptopDirector dir = new LaptopDirector(lbobj);
        Laptop lobj = dir.constructLaptop();

        System.out.println(lobj);
    }
}

```

"D:\Program Files\Java\jdk-24\bin\java.exe" "-
 Laptop [Memory=16GB, Storage=512GB]

Process finished with exit code 0

Task3

```

package Day24;

public interface Colors { 1 usage 2 implementations
    Colors clone(); 2 implementations
    String getName(); no usages 2 implementations
    void setName(String name); no usages 2 implementations
}

```

```

public class BlackConcreteprototype implements Colors { 2 ⚠️ 2 ✅ 2 ↗️
}

    public Colors clone() {
        return new BlackConcreteprototype(this.name);
    }

    public String getName() { 2 usages
        return name;
    }

    public void setName(String name) { 2 usages
        this.name = name;
    }
}

```

```

package Day24; ⚠️
public class WhiteConcretePrototype implements Colors{ 2 usages
    private String name; 4 usages
    public WhiteConcretePrototype() { no usages
        System.out.println(" WhiteConcretePrototype constructor is called");
    }
    public WhiteConcretePrototype(String name) { 2 usages
        this.name = name;
    }
    public Colors clone() {
        return new WhiteConcretePrototype(this.name);
    }
    public String getName() { 2 usages
        return name;
    }
    public void setName(String name){ 2 usages
        this.name = name;
    }
}

```

```
package Day24;
public class ClientCode {
    public static void main(String[] args) {
        Colors BlackPrototypeObj = new BlackConcretePrototype( name: "Black");
        Colors WhitePrototypeObj = new WhiteConcretePrototype( name: "White");
        Colors clonedBlackObj = BlackPrototypeObj.clone();
        Colors clonedWhiteObj = WhitePrototypeObj.clone();
        clonedBlackObj.setName("dark color");
        clonedWhiteObj.setName("light color");
        System.out.println("black color is " + clonedBlackObj.getName());
        System.out.println("while color is " + clonedWhiteObj.getName());
    }
}
```

"D:\Program Files\Java\jdk-24\b
black color is dark color
while color is light color

Process finished with exit code